

COMPSCI 689

Lecture 17: Mixture Models

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Probabilistic Unsupervised Learning

- In probabilistic unsupervised learning, our goal is to model multivariate data $\mathbf{x} = [x_1, \dots, x_D]$ generated by an unknown probabilistic process using a probabilistic model learned from a data set $\mathcal{D} = \{\mathbf{x}_n | 1 \leq n \leq N\}$.
- Since the data are vectors, we use vector-valued random variables to model them $\mathbf{X} = [X_1, \dots, X_D]$.
- Each data dimension d takes values from a potentially different set $\mathcal{X}_d$. We have $\mathbf{x} \in \mathcal{X}$. $\mathcal{X} = \mathcal{X}_1 \times \dots \times \mathcal{X}_D$.

Joint Distributions

- A probability distribution over the joint settings of multiple random variables is referred to as a *joint distribution*.
- When all dimensions of $\mathbf{x}$ are discrete, the joint distribution is represented by a *joint probability mass function*
$$P(\mathbf{X} = \mathbf{x}) = P(X_1 = x_1, \dots, X_D = x_d).$$
- When all dimensions of $\mathbf{x}$ are continuous, the joint distribution is represented by a *joint probability density function*
$$p(\mathbf{X} = \mathbf{x}) = p(X_1 = x_1, \dots, X_D = x_d).$$
- When the data are of mixed-type, we can still model them via a probability distribution consisting of both mass and density function components.

Product of Marginals

- When we need to model high-dimensional data, data of mixed types, or data that doesn't fit a simple model like a multivariate normal, we can build structured models out of different types of components.
- The most basic way to construct a structured joint probability model over a vector of random variables $\mathbf{X}$ is to model the marginal distribution of each random variable.
- The joint distribution is then defined as the product of marginals.

Product of Marginals

- Consider the case where all random variables are discrete. Let $P(X_d = x_d|\theta_d)$ be a (potentially different type of) discrete probability mass function for each dimension d . We define:

$$P(\mathbf{X} = \mathbf{x}|\theta) = \prod_{d=1}^D P(X_d = x_d|\theta_d)$$

- Consider the case where all random variables are continuous. Let $p(X_d = x_d|\theta_d)$ be a (potentially different type of) probability density function for each dimension d . We define:

$$p(\mathbf{X} = \mathbf{x}|\theta) = \prod_{d=1}^D p(X_d = x_d|\theta_d)$$

Mixed-Type Product of Marginals

- We can also specify a product of marginals over random variables of mixed types (e.g., discrete and continuous).
- Let $\mathbf{X} = [X_1, \dots, X_{D_x}]$ be a collection of continuous random variables and $\mathbf{Y} = [Y_1, \dots, Y_{D_y}]$ be a collection of discrete random variables.
- Let $P(Y_d = y_d | \theta_{yd})$ be a (potentially different type of) discrete probability mass function for each discrete variable Y_d . Let $p(X_d = x_d | \theta_{xd})$ be a (potentially different type of) probability density function for continuous dimension d . We define:

$$P(\mathbf{Y} = \mathbf{y}, \mathbf{X} = \mathbf{x} | \theta) = \prod_{d=1}^{D_y} P(Y_d = y_d | \theta_{yd}) \cdot \prod_{d=1}^{D_x} p(X_d = x_d | \theta_{xd})$$

General Mixed-Type Joint Distributions

- We can specify more general structured joint probability distributions over random variables of mixed types using the product of a conditional and marginal distribution.
- In the case shown below, the continuous random variables $\mathbf{X}$ have a probability density function that is conditioned on the discrete random variables $\mathbf{Y}$, which have a probability mass function.

$$P(\mathbf{X} = \mathbf{x}, \mathbf{Y} = \mathbf{y}) = p(\mathbf{X} = \mathbf{x} | \mathbf{Y} = \mathbf{y}) P(\mathbf{Y} = \mathbf{y})$$

- In the case shown below, the discrete random variables $\mathbf{Y}$ have a probability mass function that is conditioned on the continuous random variables $\mathbf{X}$, which have a probability density function.

$$P(\mathbf{X} = \mathbf{x}, \mathbf{Y} = \mathbf{y}) = P(\mathbf{Y} = \mathbf{y} | \mathbf{X} = \mathbf{x}) p(\mathbf{X} = \mathbf{x})$$

Inference Notes

- When we have a structured model, it is often possible to perform inference at reduced computational complexity. For example, marginalization requires no computation for a product of marginals.
- When we have a mixed type distribution, we accomplish marginalization by summing over discrete dimensions and integrating over continuous dimensions.
- When we have a mixed type distribution, conditioning is still defined via dividing the joint distribution by the marginal of the variables we are conditioning on.
- The general theory of representation and inference for structured distributions is the topic of probabilistic graphical models. The structured distributions we have introduced are special cases of Bayesian Networks.

Mixture Models

- Mixture models are universal probability distribution models constructed from simpler joint distributions over vector valued random variables $\mathbf{X}$.
- They are constructed by combining a finite discrete *latent* random variable Z with the *observed* variables $\mathbf{X}$.
- Instead of using a single product of marginals or a single multivariate normal, we use a collection of such distributions conditioned on Z .
- The joint distribution of $\mathbf{X}$ and Z is defined by:

$$P(\mathbf{X} = \mathbf{x}, Z = z | \theta) = P(Z = z | \pi) P(\mathbf{X} = \mathbf{x} | Z = z, \phi_z)$$

Mixture Models

- The distribution of $\mathbf{X}$ is given by marginalization of the joint over the values of $z \in \{1, \dots, K\}$:

$$P(\mathbf{X} = \mathbf{x}|\theta) = \sum_{z=1}^K P(Z = z|\pi)P(\mathbf{X} = \mathbf{x}|Z = z, \phi_z)$$

- The construction above is shown for an $\mathbf{X}$ that is all discrete, but the same construction works for continuous $\mathbf{X}$ as well as for a product of marginals with mixed-types.

Example: Bernoulli Mixture

Suppose the data are binary. We have $x_d \in \{0, 1\}$ for $1 \leq d \leq D$. We construct a mixture distribution for these data as follows:

$$\begin{aligned}P(Z = z|\pi) &= \pi_z \\P(X_d = x_d|Z = z, \phi_{dz}) &= \phi_{dz}^{x_d}(1 - \phi_{dz})^{1-x_d} \\P(\mathbf{X} = \mathbf{x}|Z = z, \phi_z) &= \prod_{d=1}^D P(X_d = x_d|Z = z, \phi_{dz}) \\P(\mathbf{X} = \mathbf{x}, Z = z|\pi, \phi_z) &= P(Z = z|\pi)P(\mathbf{X} = \mathbf{x}|Z = z, \phi_z) \\&= \pi_z \cdot \prod_{d=1}^D \phi_{dz}^{x_d}(1 - \phi_{dz})^{1-x_d}\end{aligned}$$

Example: Bernoulli Mixture

The mixture distribution over $\mathbf{X}$ is obtained by marginalizing the mixture indicator variable Z out of the model:

$$\begin{aligned} P(\mathbf{X} = \mathbf{x} | \pi, \phi) &= \sum_{z=1}^K P(Z = z | \pi) P(\mathbf{X} = \mathbf{x} | Z = z, \phi_z) \\ &= \sum_{z=1}^K \pi_z \cdot \prod_{d=1}^D \phi_{dz}^{x_d} (1 - \phi_{dz})^{1-x_d} \end{aligned}$$

Example: Bernoulli Mixture

Given a vector $\mathbf{x}$, we can use probabilistic inference to find the probability distribution $P(Z = z | \mathbf{X} = \mathbf{x}, \pi, \phi)$:

$$\begin{aligned} P(Z = z | \mathbf{X} = \mathbf{x}, \pi, \phi) &= \frac{P(Z = z, \mathbf{X} = \mathbf{x} | \pi, \phi)}{P(\mathbf{X} = \mathbf{x} | \pi, \phi)} \\ &= \frac{\pi_z \cdot \prod_{d=1}^D \phi_{dz}^{x_d} (1 - \phi_{dz})^{1-x_d}}{\sum_{z'=1}^K \pi_{z'} \cdot \prod_{d'=1}^D \phi_{d'z'}^{x_{d'}} (1 - \phi_{d'z'})^{1-x_{d'}}} \end{aligned}$$

If we assign each data case to the most probable value of z , this gives us a hard partitioning of the data into K clusters.

Note that when we have many dimensions, we need to be careful with this computation as both the numerator and denominator have the potential to underflow.

Example: Normal Mixture

Suppose the data x_d are real-valued for $1 \leq d \leq D$. We can model the data using a mixture where the component densities are conditional univariate normal distributions:

$$P(Z = z|\pi) = \pi_z$$

$$p(X_d = x_d|Z = z, \mu_{dz}, \sigma_{dz}^2) = \mathcal{N}(x_d; \mu_{dz}, \sigma_{dz}^2)$$

$$p(\mathbf{X} = \mathbf{x}|Z = z, \mu_z, \sigma_z) = \prod_{d=1}^D \mathcal{N}(x_d; \mu_{dz}, \sigma_{dz}^2)$$

$$\begin{aligned} P(\mathbf{X} = \mathbf{x}, Z = z|\pi, \mu, \sigma) &= P(Z = z|\pi)p(\mathbf{X} = \mathbf{x}|Z = z, \mu_z, \sigma_z) \\ &= \pi_z \cdot \prod_{d=1}^D \frac{1}{\sqrt{2\pi\sigma_{dz}^2}} \exp\left(-\frac{1}{2\sigma_{dz}^2}(x_d - \mu_{dz})^2\right) \end{aligned}$$

Example: Normal Mixture

The mixture distribution over $\mathbf{X}$ is obtained by marginalizing the mixture indicator variable Z out of the model:

$$\begin{aligned} p(\mathbf{X} = \mathbf{x} | \pi, \mu, \sigma) &= \sum_{z=1}^K P(Z = z | \pi) p(\mathbf{X} = \mathbf{x} | Z = z, \mu_z, \sigma_z) \\ &= \sum_{z=1}^K \pi_z \cdot \prod_{d=1}^D \frac{1}{\sqrt{2\pi\sigma_{dz}^2}} \exp\left(-\frac{1}{2\sigma_{dz}^2}(x_d - \mu_{dz})^2\right) \end{aligned}$$

Example: Normal Mixture

Given a vector $\mathbf{x}$, we can use probabilistic inference to infer the probability distribution over Z :

$$\begin{aligned} P(Z = z | \mathbf{X} = \mathbf{x}, \pi, \mu, \sigma) &= \frac{P(Z = z, \mathbf{X} = \mathbf{x} | \pi, \mu, \sigma)}{p(\mathbf{X} = \mathbf{x} | \pi, \mu, \sigma)} \\ &= \frac{\pi_z \cdot \prod_{d=1}^D \frac{1}{\sqrt{2\pi\sigma_{dz}^2}} \exp\left(-\frac{1}{2\sigma_{dz}^2} (x_d - \mu_{dz})^2\right)}{\sum_{z'=1}^K \pi_{z'} \cdot \prod_{d'=1}^D \frac{1}{\sqrt{2\pi\sigma_{d'z'}^2}} \exp\left(-\frac{1}{2\sigma_{d'z'}^2} (x_{d'} - \mu_{d'z'})^2\right)} \end{aligned}$$

Note that when we have many dimensions, we need to be careful with this computation as both the numerator and denominator have the potential to underflow or overflow.

Example: Multivariate Normal Mixture

Suppose the data x_d are real-valued for $1 \leq d \leq D$. We can model the data using a mixture where the component densities are conditional multivariate normal distributions:

$$P(Z = z|\pi) = \pi_z$$

$$p(\mathbf{X} = \mathbf{x}|Z = z, \mu_z, \Sigma_z) = \frac{1}{|2\pi\Sigma_z|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_z)^T \Sigma_z^{-1}(\mathbf{x} - \mu_z)\right)$$

$$\begin{aligned} P(\mathbf{X} = \mathbf{x}, Z = z|\pi, \mu, \sigma) &= P(Z = z|\pi)p(\mathbf{X} = \mathbf{x}|Z = z, \mu_z, \Sigma_z) \\ &= \pi_z \cdot \frac{1}{|2\pi\Sigma_z|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_z)^T \Sigma_z^{-1}(\mathbf{x} - \mu_z)\right) \end{aligned}$$

Example: Multivariate Normal Mixture

The mixture distribution over $\mathbf{X}$ is obtained by marginalizing the mixture indicator variable Z out of the model:

$$\begin{aligned} p(\mathbf{X} = \mathbf{x} | \pi, \mu, \sigma) &= \sum_{z=1}^K P(Z = z | \pi) p(\mathbf{x} = \mathbf{x} | Z = z, \mu_z, \sigma_z) \\ &= \sum_{z=1}^K \pi_z \cdot \frac{1}{|2\pi\Sigma_z|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_z)^T \Sigma_z^{-1}(\mathbf{x} - \mu_z)\right) \end{aligned}$$

Example: Multivariate Normal Mixture

Given a vector $\mathbf{x}$, we can use probabilistic inference to infer the probability distribution over Z :

$$\begin{aligned} P(Z = z | \mathbf{X} = \mathbf{x}, \pi, \mu, \sigma) &= \frac{P(Z = z, \mathbf{X} = \mathbf{x} | \pi, \mu, \sigma)}{p(\mathbf{X} = \mathbf{x} | \pi, \mu, \sigma)} \\ &= \frac{\pi_z \cdot \frac{1}{|2\pi\Sigma_z|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_z)^T \Sigma_z^{-1} (\mathbf{x} - \mu_z)\right)}{\sum_{z'=1}^K \pi_{z'} \cdot \frac{1}{|2\pi\Sigma_{z'}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu_{z'})^T \Sigma_{z'}^{-1} (\mathbf{x} - \mu_{z'})\right)} \end{aligned}$$

Note that when we have many dimensions, we need to be careful with this computation as both the numerator and denominator have the potential to underflow or overflow.

Learning for Mixture Models

- As probabilistic models, mixture models are most often learned using maximum likelihood based criteria.
- In a mixture model, the full joint distribution includes the data variables $\mathbf{X}$ and the latent mixture indicator variable Z .
- Since the mixture indicator variables Z are not observed, we need to marginalize them out of the model and then minimize the negative log likelihood of the marginalized distribution.
- We refer to the resulting optimization criterion as the *negative log marginal likelihood* (NLML) function.

Learning for Mixture Models

The negative log marginal likelihood for a generic mixture model is given below:

$$\begin{aligned}nlml(\mathcal{D}, \theta) &= - \sum_{n=1}^N \log p(\mathbf{x}_n | \theta) \\ &= - \sum_{n=1}^N \log \left(\sum_{z=1}^K P(Z = z | \pi) p(\mathbf{X}_n = \mathbf{x}_n | Z = z, \theta) \right)\end{aligned}$$

We can learn mixture models using direct negative log marginal likelihood minimization. Note that parameter transformations must be used to deal with constrained parameter spaces.

Example: NLML for Mixture of Bernoullis

$$P(\mathbf{X} = \mathbf{x}, Z = z | \pi, \phi_z) = P(Z = z | \pi) P(\mathbf{X} = \mathbf{x} | Z = z, \phi_z)$$

$$= \pi_z \cdot \prod_{d=1}^D \phi_{dz}^{x_d} (1 - \phi_{dz})^{(1-x_d)}$$

$$P(\mathbf{X} = \mathbf{x} | \pi, \phi_z) = \sum_{z=1}^K \pi_z \cdot \prod_{d=1}^D \phi_{dz}^{x_d} (1 - \phi_{dz})^{(1-x_d)}$$

$$nlml(\mathcal{D}, \theta) = - \sum_{n=1}^N \log \left(\sum_{z=1}^K \pi_z \cdot \prod_{d=1}^D \phi_{dz}^{x_{nd}} (1 - \phi_{dz})^{(1-x_{nd})} \right)$$

Example: NLML Minimization for Mixture of Bernoullis

$$\begin{aligned}\hat{\theta} &= \arg \min_{\theta} - \sum_{n=1}^N \log \left(\sum_{z=1}^K \pi_z \cdot \prod_{d=1}^D \phi_{dz}^{x_{nd}} (1 - \phi_{dz})^{(1-x_{nd})} \right) \\ \text{s.t., } \forall d, z, \quad &0 \leq \phi_{dz} \leq 1 \\ &\forall z, \pi_z > 0 \\ &\sum_{z=1}^K \pi_z = 1\end{aligned}$$

Example: NLML for Mixture of Normals

$$P(\mathbf{X} = \mathbf{x}, Z = z | \pi, \phi_z) = P(Z = z | \pi) p(\mathbf{X} = \mathbf{x} | Z = z, \phi_z)$$

$$= \pi_z \cdot \prod_{d=1}^D \mathcal{N}(x_d; \mu_{dz}, \sigma_{dz}^2)$$

$$P(\mathbf{X} = \mathbf{x} | \pi, \phi_z) = \sum_{z=1}^K \pi_z \cdot \prod_{d=1}^D \mathcal{N}(x_d; \mu_{dz}, \sigma_{dz}^2)$$

$$nlml(\mathcal{D}, \theta) = - \sum_{n=1}^N \log \left(\sum_{z=1}^K \pi_z \cdot \prod_{d=1}^D \mathcal{N}(x_d; \mu_{dz}, \sigma_{dz}^2) \right)$$

Example: NLML Minimization for Mixture of Normals

$$\begin{aligned}\hat{\theta} &= \arg \min_{\theta} - \sum_{n=1}^N \log \left(\sum_{z=1}^K \pi_z \cdot \prod_{d=1}^D \mathcal{N}(x_d; \mu_{dz}, \sigma_{dz}^2) \right) \\ \text{s.t., } &\forall d, z, \sigma_{dz} > 0 \\ &\forall z, \pi_z > 0 \\ &\sum_{z=1}^K \pi_z = 1\end{aligned}$$